

PIPELINE REPORT

How DeciWaves works

DeciWaves

DeciWaves turns your own installed copies of three games (Death Stranding, Horizon Zero Dawn Remastered, Horizon Forbidden West) into a spreadsheet of every spoken line and a set of story-ordered MP3 files. You point it at a game you own; it reads, and never changes, the game's files, works out which recorded clip carries which subtitled line and who said it, puts those lines in story order, and stitches them into MP3 reels of about 290 MB each with a tracklist beside every one. Windows only. It ships no game content of its own.

Covers all three games. They share one skeleton but the middle differs per game, so the report follows the shared five stages and notes where each game diverges. This supersedes the report dated 15 July 2026: every fragility that one listed has since been fixed in the code, so what follows describes the pipeline as it stands on 17 July 2026, and what still deserves attention. Where this and the code disagree, the code is right.

Generated 17 July 2026.

Read the code, not just this. Where the two disagree, the code is right.

1.

What this is

DeciWaves is a command-line tool for someone who owns these three games and wants the voice acting as listenable audio, indexed and in story order. The unit of work is one game. For each one, “done” means two things: a manifest, a CSV spreadsheet with one row per voice line carrying a stable id, the line’s internal name, its speaker where that can be derived, its scene and category, its language, and a path to a playable clip; and a set of MP3 reels assembled from those clips in narrative order.

Three rules shape everything. It is read-only against any install it touches, so it never repacks or edits game files. It ships code, never game content, so copyrighted script text is something you supply yourself and point the tool at. And it is Windows only, because it shells out to Windows decoder binaries for the audio.

The three games look similar from the outside and are genuinely different underneath. Death Stranding stores dialogue in Decima engine resources that name each line’s audio directly. The two Horizon games use newer, different archive formats that do not, so most of the engineering lives in working out which recorded clip goes with which line. Treat every format detail here as reverse engineered and likely, not certified: it comes from inspecting files, not from documentation.

2.

At a glance

- 1 Configure and check**
Point it at a game you own; fetch the decoder tools; confirm what is present.
- ↓
- 2 Catalog**
Read the install and list every dialogue line: id, name, speaker, subtitle, scene.
- ↓
- 3 Bind**
Attach the actual recorded clip to each listed line. The hard part; each game solves it differently.
- ↓
- 4 Order**
Arrange the bound lines in story sequence.
- ↓
- 5 Render**
Decode the clips and pack them into MP3 reels, each with a tracklist.

The five stages are conceptual. Death Stranding skips Bind entirely, because its catalog already names each line's audio. The Horizon games fold Order into their manifest rather than running it as a separate step. The command that runs a game end to end (`deciwaves <game> run`) executes only the stages that game actually has, and skips any whose work is already done.

3.

Stage by stage

Configure and check readiness

`deciwaves setup` fetches `ffmpeg`, `vgmstream-cli` and `VGAudioCli` into `%LOCALAPPDATA%\DeciWaves\tools` and records where each game and the Oodle DLL live. It merges rather than overwrites: an omitted flag keeps the value you saved before, and passing an empty string clears one, so registering a second game never blanks the first game's paths. A corrupt or hand-edited `config.json` is ignored with a warning and the run starts from an empty config, rather than crashing every command.

`deciwaves doctor` prints a readiness list: each decoder tool present or missing with the exact fix, each game found or not configured, and, as information only, whether the `asr` extra and a CUDA GPU are visible. A game you do not own reports “not configured” and never counts as a failure.

IN	an internet connection; the game or games you own
DOES	downloads the decoder tools and records where things live
OUT	a saved <code>config.json</code> plus the tools on disk
NEEDS	<code>github.com</code> reachable; for the Horizon games, a CUDA GPU and the <code>asr</code> extra
FILES	<code>cli/setup.py</code> <code>cli/doctor.py</code> <code>cli/config.py</code>

Catalog: list every line

Death Stranding reads the Decima sentence resources directly and records each line's id, internal name, category, scene, speaker and English subtitle, alongside the path to its audio. Horizon Zero Dawn cannot do this: its archive exposes only path hashes, not names, so the catalog recovers sentence-core paths by scanning file bodies for readable path strings, then reads the same fields. Forbidden West takes a third route, extracting every provable English dialogue clip straight to a WAV file up front.

The catalog columns are `line_id`, `core_path`, `line_index`, `category`, `scene`, `speaker_code`, `speaker_name`, `subtitle_en`, `wem_path_en` and `language`. There is no per-line length or duration column. How long a clip runs is not known here; it is measured later, by decoding the clip during render. Anything showing a running time before render would have to decode the clip itself to get it.

Resume is deliberate. A line's row counts as finished only once a separate sidecar file confirms its whole source core completed, and a startup pass drops any rows left behind by a source that crashed partway. That is what stops a half-processed file from looking done and quietly losing the rest of its lines on the next run.

IN	the game install
DOES	enumerates dialogue lines and their metadata into a CSV
OUT	<code>catalog.csv</code> (DS, HZD) or <code>clip-index.csv</code> (FW), plus a resume sidecar
NEEDS	the Oodle DLL (DS); nothing external for HZD's scan; VGAudio for FW's decode
FILES	<code>games/ds/catalog.py</code> <code>games/hzd/catalog.py</code> <code>games/fw/extract.py</code> <code>engine/catalog_io.py</code>

Bind: attach a clip to each line

This is where the three games stop resembling each other, and it is the reason the project is really three solutions sharing a frame. Death Stranding has no bind step at all: the catalog already names each line's audio, so there is nothing to match, and cutscene voice comes from whole-scene tracks resolved once and shipped with the tool.

Horizon Zero Dawn starts with lines and clips that are both anonymous. Each is fingerprinted by two cheap numbers, its encoded byte length and its sample count. A fingerprint that matches exactly one line to exactly one clip is bound immediately with no further work, marked tier S. Fingerprints shared by several lines and clips are ambiguous, and only those are handed to a local speech-to-text model and matched by their text.

Forbidden West transcribes every clip with the same local model, assigns the game's exact on-screen subtitle to each clip (using the transcript only to pick which clip within a group), then matches each subtitled line against your own copy of the game's script. The script is the story spine: a line with no home in the script is a bark and never binds, and the script also supplies the speaker and the running order. The Horizon binds can run for hours on a full library.

The `tier` field records how a row was bound, not what kind of line it is. Tier S is a fingerprint bind; other tiers mark subtitle, woven and DLC binds. Render later selects rows by tier. It is not a bark-versus-dialogue flag, and there is no separate field that is.

IN	the catalog plus the game's raw audio clips
DOES	decides which recorded clip carries which line
OUT	a bind manifest, with a <code>tier</code> on each row recording how it was matched
NEEDS	Death Stranding, nothing; Horizon games, a CUDA GPU and the <code>asr</code> extra
FILES	<code>games/hzd/binding.py</code> <code>games/hzd/asr_bind.py</code> <code>games/fw/subtitle_bind.py</code> <code>games/fw/subtitle_match.py</code>

Order: put the lines in story sequence

Death Stranding has a real, separate order step. It keeps only lines that have a subtitle, removes within-scene duplicates, and places scenes in narrative order, anchored to a transcript you supply where one covers a scene and falling back to built-in heuristics elsewhere. The result is `playlist.csv`, tagged so that cutscene and mission lines form the spine and everything else is side content.

The Horizon games have no standalone order command. Their order is a byproduct of the bind: Horizon Zero Dawn uses quest numbering, Forbidden West uses the position of each line in your script. That order is already written into the bind manifest, so there is no separate marker or file for it.

IN	the catalog (DS) or the bind manifest (Horizon)
DOES	selects and orders lines into a playable sequence
OUT	<code>playlist.csv</code> (DS); order baked into the manifest (Horizon)
NEEDS	an optional narrative transcript (DS) for tighter placement
FILES	<code>games/ds/story_order.py</code> <code>engine/selection.py</code> <code>engine/transcript_anchor.py</code>

Render: pack into MP3 reels

Render decodes each clip to WAV, caching as it goes so re-runs are cheap, drops a short silence between lines and a longer one between scenes, and concatenates the result into MP3 files. It sizes each file to land just under 290 MB, using a fixed target near 285 MB and the chosen bitrate. The size ceiling itself is not a flag; the bitrate (`--bitrate`) is the only lever, and lower bitrates simply mean fewer, larger reels.

Selecting a subset does have a supported answer today, contrary to what earlier notes assumed. Render reads its lines from a playlist or manifest path you can point elsewhere, and it filters rows by their tier (Forbidden West exposes this as `--tiers`). Death Stranding also has `--main-story`, which keeps only the narrative spine. So rendering a chosen set of lines means handing render a filtered playlist or manifest in its own format, not passing a per-line argument. Each reel gets a tracklist sidecar carrying the timestamp, scene or quest, speaker, subtitle and line id.

Render will not fake success. If it attempted to decode clips and none of them worked, it stops with an error that points at `doctor` and the per-clip error log, rather than writing a silent, empty reel.

IN	the ordered playlist or manifest
DOES	decodes each clip and concatenates the clips into sized MP3 files
OUT	<code>phase_d_NN.mp3</code> plus <code>phase_d_NN.tracklist.csv</code> per reel
NEEDS	ffmpeg, plus the game's own decoder (vgmstream and Oodle, or VGAudio)
FILES	<code>engine/render.py</code> <code>games/hzd/render.py</code> <code>games/fw/render.py</code> <code>engine/audio_clip.py</code>

4.**The map**

<code>src/deciwaves/engine/</code>	The shared, game-agnostic core: the render, selection and resume-bookkeeping code every game reuses, plus decoder-path resolution.
<code>src/deciwaves/engine/pack/</code>	The format readers, one family per game: DS archives, the HZD package, the FW streaming graph.
<code>src/deciwaves/games/ds hzd fw/</code>	Each game's own catalog and bind code, and Death Stranding's own order step. This is where the three approaches actually live.
<code>src/deciwaves/cli/</code>	The commands: <code>main</code> dispatches, <code>run</code> chains a game end to end, <code>setup</code> and <code>doctor</code> handle readiness, <code>guided</code> is the menu.
<code>src/deciwaves/_vendor/pydecima/</code>	A bundled, patched Decima resource parser, used by Death Stranding only.
<code>src/deciwaves/data/</code>	Small reference files shipped with the tool: file lists, pre-resolved cutscene tracks, character rosters.

out/ (per workspace)	Everything a run produces: the CSVs, the WAV cache, the MP3 reels, and the <code>.done-<stage></code> resume markers. Kept out of git.
.memories/	Durable notes on the reverse-engineered formats. The best place to understand why a game needed the solution it got.

5.

How to run it

1. Install the package

```
pip install -e ".[test]" # add [asr] as well for the Horizon games
```

The base install runs the tests and the DS pipeline. The Horizon binds need the asr extra plus a CUDA-matched PyTorch.

2. Fetch tools and register a game

```
deciwaves setup --ds-install "D:\ ... \DEATH STRANDING DIRECTOR'S CUT"
deciwaves doctor
```

HZD wants the `...\LocalCacheDX12\package` dir (the one with `PackFileLocators.bin`); FW wants `...\LocalCacheWinGame\package` (with `streaming_graph.core`). `doctor` tells you what is still missing.

3. Run a game end to end

```
deciwaves --workspace D:\deciwaves ds run
deciwaves --workspace D:\deciwaves hzd run
deciwaves --workspace D:\deciwaves fw run --gamescript <your-script>
```

The `--workspace` flag MUST come before the game name. Forbidden West produces subtitle-labelled reels without a script, but needs your own script for speakers and story order.

4. Collect the output

```
dir D:\deciwaves\out\ds
```

The manifest CSV and the `phase_d_NN.mp3` reels with their tracklists land under `out\<game>`.

6.

What could bite you

The classic pipeline hazards are already handled here, and it is worth saying so plainly. Files are written to a temporary name and moved into place, so a crash never leaves a half-written file looking valid. Resume state has one authority, so a re-run cannot silently lose work. Re-running an early stage automatically invalidates the later stages' markers, so stale downstream output cannot pass for fresh. The missing-GPU case is caught before any hours-long stage starts, not after. What remains is smaller and mostly about visibility.

A capped Horizon bind looks identical to a complete one, on disk

MEDIUM

FIX: S

`src/deciwaves/games/hzd/asr_bind.py`, lines 143–152

- RISK** `hzd run` caps how many ambiguous clips get transcribed at 300 by default. When the cap trips, bind prints how many buckets it skipped, but writes nothing to disk about it; the skipped lines are simply absent from the manifest and the reels. Once that console line scrolls away, nothing reading the CSVs, a coverage check, a script, a future front-end, can tell a capped rip from a complete one.
- FIX** Record the skipped-bucket count and ids to a small sidecar next to the manifest, so completeness is a fact on disk and not just a line in the log. Run with `--sample-cap 0` for a full, uncapped pass.

Horizon catalog drops cores it cannot read, without a trace

MEDIUM

FIX: S

`src/deciwaves/games/hzd/inventory.py`, lines 43–46

- RISK** The path-harvest that runs on every Horizon Zero Dawn catalog swallows a read failure on any core and moves on, with no log line and no counter. A sentence core that fails to read simply vanishes, the lines it held never appear in the catalog, and nothing tells you how many were lost. Every sibling step (the DS and HZD catalogs, the metadata pass, the FW extract, the render loop) logs its failures to an error file. This one alone does not.
- FIX** Count the failures and write them to an errors file, exactly as the neighbouring stages already do. The fix is a few lines and makes a silent gap into a visible one.

Re-running a Horizon bind with a bigger cap does nothing until you delete the marker

LOW

FIX: M

`src/deciwaves/cli/run.py`, lines 305–310

- RISK** Resume keys purely on whether the `.done-bind` marker exists. If a first run capped the transcription and you run again with a larger `--sample-cap` to get a fuller pass, the stage is skipped as already done and nothing new happens. The behaviour is spelled out in `--help`, so it is documented rather than hidden, but it still surprises anyone who does not read it.
- FIX** Delete `out/hzd/.done-bind` and re-run to force the fuller pass. A sturdier fix would record the cap the marker ran with and re-run automatically when it changes.

A misplaced `--workspace` fails with a message that blames the wrong thing

LOW

FIX: S

`src/deciwaves/cli/main.py`, lines 100–113

- RISK** `--workspace` is a global flag and must come before the game name. Put it after (`deciwaves ds --workspace D run`) and it is swallowed as part of the stage's own arguments; the run stops with "invalid choice: '–workspace'", which points at the stage name rather than at the real cause. It fails loudly, so no wrong output is produced, but the error misdirects.
- FIX** Keep `--workspace` before the game name. A clearer error when a flag-shaped token lands in the stage position would save the guesswork.

7.

Open questions

? Forbidden West ships three ways to assemble a reel: the full reel, a woven variant, and a separate DLC path. The end-to-end `fw run` uses only the full reel. Whether the other two are finished features worth exposing or exploratory leftovers is not decidable from the code alone.

? The Death Stranding `cutscenes` stage is not in the default chain; the chain uses a pre-resolved cutscene-track file shipped with the tool. Whether users are expected to regenerate it against a patched install, or treat the shipped file as canonical, is a product decision the code does not settle.

8.

Glossary

Decima The game engine all three games are built on. DeciWaves reads its data files; it does not run the engine.

sentence core A Decima data record describing one line of dialogue: its text, speaker, and a reference to its audio.

Wwise / .wem The audio format the recorded voice sits in inside Death Stranding. Decoding it needs a Wwise-aware tool, `vgmstream`.

ATRAC9 The compressed audio format the two Horizon games use for their clips, decoded with `VGAudio`.

Oodle A compression library Death Stranding uses. Its DLL ships next to the game's executable, and DeciWaves needs it to open the archives.

fingerprint Two cheap numbers, a clip's encoded byte length and its sample count, used to tell clips apart without decoding them.

structural bind A line-to-clip match made from fingerprints alone, with no speech-to-text. Marked tier S.

speech-to-text (ASR) Turning recorded speech into text, run locally on the GPU. Used only to disambiguate clips a fingerprint cannot separate.

gamescript The game's own script, in play order. A file you supply yourself; it gives Forbidden West its speakers and story order.

bark A short, unscripted incidental line, a guard's shout or an idle mutter, as opposed to scripted story dialogue.

tier A tag on each bound line recording how it was matched. Render selects which tiers to include. Not a line-type label.

resume marker An empty `.done-<stage>` file recording that a stage finished, so a re-run skips it.

reel One packed MP3 file of story-ordered lines, kept just under 290 MB, with a tracklist beside it.